

AIGUL SHARIP

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PROFILE

Bioinformatician & Software Engineer with expertise in multi-omics analysis, machine learning, and large-scale biomedical data integration. Published over 10 peer-reviewed papers, including five as first author, with research spanning genomics, transcriptomics, and systems biology. Motivated to apply computational approaches to integrative biomedical data analysis, uncover novel disease mechanisms, and advance precision medicine for improved diagnostics and treatment strategies.

WORK EXPERIENCE

Software engineer/ Bioinformatician,

Mar 2025 – Present

EPAM Systems is a leading global IT consultancy with offices in 50+ countries. Invitae is a clinical genomics company providing genetic testing and precision medicine solutions.

- Contributed to the development of Genosity [Integrated Genomics Toolkit](#), a large-scale platform for managing and analyzing NGS data.
- Designed and implemented bioinformatics workflows and pipelines supporting clinical genomics and precision medicine applications.
- Collaborated with cross-functional teams to align computational solutions with bioinformatics workflows and clinical data processing needs.
- Support backend development using Java (Spring Boot) and optimized infrastructure with AWS
- *Technical Stack: Java, Spring Boot, Nextflow, Python, R, AWS, Docker, Gitlab, Unix/Linux*

Bioinformatician, part-time

Mar 2019 – Present

Lab. of Bioinformatics and System Biology, Nazarbayev University

Astana, Kazakhstan

- Developed machine learning models for Alzheimer's disease prediction (cognitive scores and biomedical features) and built a prototype Shiny web [AD Prediction app](#) enabling rapid AD risk assessment.
- Performed large-scale WGS and RNA-seq analysis of Kazakhstani individuals, identifying disease-associated variants and pathways
- Conducted statistical analysis of risk factors of the severity of coronary atherosclerosis from patient data
- Carried out genome-wide association studies (GWAS) to identify genetic variants linked to complex diseases and traits
- Co-authored 10+ research publications, 7+ international internships and conferences (USA, Germany, Japan)
- *Technical Stack: R, Unix/Linux, Python, multi-omics data analysis, biostatistics, ML models*

Software Engineer

Mar - Sep 2024

BTS Digital

Astana, Kazakhstan

- Contributed to the development of the *Ismet platform* for Kazakhtelecom Business users.
- Developed *Ismet Docs* platform to enhance electronic document management.
- *Technical Stack: Java, Spring, SQL, JavaScript, Postgres, GitLab, MinIO, Jasper, Agile*

Software Engineer

Jul 2022 – Jan 2024

Datanomix/Motio

Dallas, Texas, USA

- Participated in the enhancement of [Soterre](#), version control and deployment tool, designed for Qlik users.
- Executed the integration of external Qlik Sense platform functionalities for more user-friendly experience
- *Technical Stack: Java, Spring, JavaScript, ExtJS, SQL, Postgres, Tomcat, Docker, GitLab, Agile*

Full-stack Software Engineer participant

Jun-Aug 2022

nFactorial Incubator, 10-week web development bootcamp

Almaty, Kazakhstan

- Developed full-stack web app [Recipe Generator](#) for finding recipes with various functionalities
- *Technical Stack: React, JavaScript, JQuery, REST API, Bootstrap*

Research Assistant

Jan 2017 - Mar 2019

Lab of Dr. Jeanette Kunz, Nazarbayev University

Astana, Kazakhstan

- Contributed to the delivery of the research project "Phosphoinositide phosphate signaling networks: A novel therapeutic opportunity in breast cancer"
- Performed molecular biology experiments: design PCR primers, PCR, transfection, cell culturing, microscopy, image analysis, FACS, DNA isolation etc.

EDUCATION

Nazarbayev University (NU)

2017–2019

- Master of Biological Sciences, **GPA 4.0/4.0**, with the Highest distinction
- Honors: Dean's list for Spring & Fall 2018; **Received Honor Diploma from the President of Kazakhstan**

Nazarbayev University

2013–2017

- Bachelor of Biological Sciences, GPA 3.66/4.0
- Honors: Dean's list for Fall 2015 and Fall 2017 semesters

TECHNICAL SKILLS

Software Engineering	Java, JavaScript, R, Python, Nextflow, Unix/Linux, CSS, SQL, Postgres, React
ML & DS skills	supervised/unsupervised ML, feature selection, predictive modeling, statistics
Bioinformatics	multi-omics data analysis, WGS, RNA-seq, GWAS data analysis
Frameworks	Spring Framework, Java EE, Git, REST API, JUnit, Postman, Docker, Kubernetes, Microservices, OpenFeign, Eureka, Apache Kafka, Elasticsearch Agile
Languages	Fluent in Kazakh, Russian and English, Intermediate German, Basic French

EXTRACURRICULUM ACHIEVEMENTS

- Received 10+ PhD offers (ETH Zurich/Uni. of Zurich, INPhINIT fellowship, DKFZ, Uni. of Copenhagen, 3 Marie-Curie fellowships, Institut Curie etc.)
- Awarded comprehensive fellowship to participate the Yessenov Data Lab program, summer 2021
- Travel award from School of Science and Technology, NU for internship at DRFZ, Germany, July-August 2018
- Scholarship for Research Internship from IRCMS, Kumamoto University, Japan, Nov-Dec 2017
- Aegean Conference at the 8th International Conference on Autoimmunity, Rhodes, Greece, Sep 2017
- Third prize in the Excellent Article Award in the 5th Guangzhou International Symposium., China, Dec 2018
- Winner of Yessenov Foundation Summer Research Fellowship for internship at UCSF, USA, Jan 2016
- Silver medal for NU Biology team at International iGEM competition, Boston, USA, Sep 2015

PUBLICATIONS

1. Serikzhan A., **Sharip, A.**, Daniyarov A., ... & Kairov U. (2025). Genomic landscape of the Great Steppe: Genetic variants in healthy Kazakh individuals. *Scientific Data*, accepted for publication on 26 August.
2. **Sharip, A.**, & Kunz, J. (2025). Mechanosignaling via Integrins: Pivotal Players in Liver Fibrosis Progression and Therapy. *Cells* 14(4), 266
3. **Sharip, A.**, Rakhimova, S., Molkenov, A., Ashenova, A., Kozhamkulov, U., Akhmetollayev, I., ... & Kairov, U. (2024). Transcriptome profiling and analysis of patients with esophageal squamous cell carcinoma from Kazakhstan. *Frontiers in Genetics*, 15, 1249751.
3. Kairov, U.*, Molkenov, A*., **Sharip, A.***, Rakhimova, S., Seidualy, M., Rhie, A., ... & Akilzhanova, A. (2022). Whole-genome sequencing and genomic variant analysis of Kazakh individuals. *Frontiers in Genetics*, 13, 902804. * Equal contribution
4. **Sharip, A.**, Mukhatayev, Z., Chunikhina, D., Baglanova, M., Poddighe, D., Ainabekova, B., ... & Kunz, J. (2022). Microorganisms in the Pathogenesis and Management of Ankylosing Spondylitis. In *Role of Microorganisms in Pathogenesis and Management of Autoimmune Diseases: Volume I: Liver, Skin, Thyroid, Rheumatic & Myopathic Diseases* (pp. 459-487). Singapore: Springer Nature Singapore.
5. Mukhatayev, Z., **Sharip, A.**, Nurgozhina, A., Chunikhina, D., Poddighe, D., Ainabekova, B., ... & Kunz, J. (2022). Microorganisms in the Pathogenesis and Management of Spondyloarthritis. *Role of Microorganisms in Pathogenesis and Management of Autoimmune Diseases: Volume I: Liver, Skin, Thyroid, Rheumatic & Myopathic Diseases*, 419-458.
6. Seisenova, A., Daniyarov, A., Molkenov, A., **Sharip, A.**, Zinovyev, A., Kairov, U. Meta-Analysis of Esophageal Cancer Transcriptomes Using Independent Component Analysis. *Frontiers in Genetics* 2021, 12
7. Karabayev, D., Molkenov, A., Yerulanuly, K., Kabimoldayev, I., Daniyarov, A., **Sharip, A.**, Seisenova, A.; Zhumadilov, Z.; Kairov, U. re-Searcher: GUI-based bioinformatics tool for simplified genomics data mining of VCF files. *PeerJ* 2021, 9, e11333,
8. Kairov, U., Molkenov, A., Rakhimova, S., Kozhamkulov, U., **Sharip, A.**, Karabayev, D., Daniyarov, A., H.Lee, J., D.Terwilliger, J., Akilzhanova, A., Zhumadilov, Z. Whole-genome sequencing data of Kazakh individuals. *BMC Research Notes* 2021, 14, 45
9. **Sharip, A.**, & Kunz, J. (2020). Understanding the pathogenesis of spondyloarthritis. *Biomolecules*, 10(10), 1461.
10. Sharip, A., Abdukhakimova, D., Wang, X., Kim, A., Kim, Y., **Sharip, A.**, ... & Xie, Y. (2017). Analysis of origin and protein-protein interaction maps suggests distinct oncogenic role of nuclear EGFR during cancer evolution. *Journal of Cancer*, 8(5), 903.

11. Kim, Y., Kim, A., Sharip, A., **Sharip, A.**, Jiang, J., Yang, Q., & Xie, Y. (2017). Reverse the resistance to PARP inhibitors. *International journal of biological sciences*, 13(2), 198.

RESEARCH EXPERIENCE

- Bioinformatician at National Laboratory Astana, NU** Sep 2018 - Present
Prof. Dr. Ulykbek Kairov, Lab. of Bioinformatics and System Biology, NU Astana, Kazakhstan
- conducted whole-genome sequencing and variant analysis of Kazakh individuals, identifying genetic variants linked to metabolic and neurodegenerative diseases
 - performed transcriptome profiling of Kazakhstani ESCC patients, discovering key dysregulated pathways and potential biomarkers for early diagnosis and precision therapies.
 - analyzed statistical risk factors influencing the severity of coronary atherosclerosis from patient data
- Laboratory research for Master Thesis “The role of bacteria and HLA-B27*05 allele on the pathogenesis of spondylarthritis”** Aug 2017 - May 2019
Prof. Jeannette Kunz and Prof. Christian Schoenbach, NU Astana, Kazakhstan
- immunoinformatics analysis of potential Brucella epitopes, cell-based epitope binding assays and FACS analysis of gut microbiota
- Research internship at German Rheumatism Research Center (DKFZ)** Jul - Aug 2018
Lab. of Dr. Kruglov, DKFZ Berlin, Germany
- analysis of gut microbiota of AS patients and healthy controls by staining and sorting of flow cytometry DB Influx, stimulation experiments with human PBMC cells
- Research internship at Kumamoto University** Nov-Dec 2017
Prof. Christian Schoenbach, IRCSM, Kumamoto University Kumamoto, Japan
- analysis of gut microbiota of AS patients and healthy controls by staining and sorting of flow cytometry DB Influx, stimulation experiments with human PBMC cells
- Laboratory research for Bachelor Thesis “Regulation of Focal Adhesion Assembly and Lamellopodia Dynamics with WNT Signalling Ligands”** Aug 2016 - May 2017
Prof. Adarichev and Prof. Jeannette Kunz, NU Astana, Kazakhstan
- cell culturing and analysis rheumatoid arthritis cell lines with immunofluorescence techniques, optimize high-resolution time-lapse microscopy, plasmid isolation
- Research internship at University of California San Francisco (USCF)** May - Aug 2016
Lab. of Prof. Rosemary Akhurst, Helen Diller Comprehensive Cancer Center, USCF San Francisco, USA
- gene cloning by expression vectors, restriction enzyme digestions, DNA extraction from tissue and genotyping, primary cell culturing, siRNA transfection, Western blot

PROJECTS & COURSES

- Software Engineer training** Nov 2020 – Aug 2021
nFactorial School
- Completed nFactorial Start - Introduction to Computer Science, 8-week intensive program in Java, with **Gold Certificate of Achievement**, accomplished 11 Java projects
 - Improved problem-solving skills at 8-week intensive nFactorial Algorithms and Data Structures course
- Java Backend Developer training** Oct 2021 – Mar 2022
BitLab Academy & OneLab
- Acquired a solid understanding of front-end development essentials using HTML, CSS, and JavaScript
 - Explored the core principles of back-end development utilizing Java EE and Spring Frameworks
 - Engineered an **Online Pharmacy** web application leveraging Spring Boot, Spring Data, and Spring Security
- Data Scientist training** Jun–Aug 2021
Yessonov Data Lab, 8-week summer school for Data Scientist
- Acquired skills in data preprocessing, visualization and interpretation with Python and R
 - Covered the fundamentals of machine learning, neural networks
 - Mastered the techniques of statistical analysis and modelling
- Certifications from completed courses:** <https://github.com/aigulsharip/certificates>